

Diabetic Retinopathy Severity Classification Using Deep Learning

Abstract

1 Introduction

2 Problem Statement

3 Methodology

As discussed previously, the baseline model’s performance is significantly constrained by two major limitations in the dataset and problem structure: severe class imbalance and the ordinal nature of diabetic retinopathy severity grades. The dataset is heavily skewed toward lower-grade samples, causing the model to bias predictions toward majority classes while underperforming on minority classes representing more severe disease stages. In addition, diabetic retinopathy classification is inherently ordinal, meaning that misclassifications should not all be treated equally. For example, predicting a Grade 4 image as Grade 3 is considerably less severe than predicting it as Grade 0, yet standard Cross Entropy Loss penalizes both errors similarly because it treats classes as independent categories rather than ordered progression levels. This limits the model’s ability to learn the structured severity relationship between grades.

To address these shortcomings, we propose the use of a hybrid loss function that combines Focal Loss and CORAL (Consistent Rank Logits) Loss. Focal Loss is specifically designed to mitigate class imbalance by dynamically down-weighting easy, high-confidence predictions and assigning greater emphasis to hard or misclassified examples. This will ensure that minority and difficult samples contribute more significantly during backpropagation, improving sensitivity to underrepresented disease stages. CORAL Loss, on the other hand, is designed for ordinal classification problems by transforming multi-class prediction into a series of ordered binary threshold tasks. Instead of treating grades as unrelated labels, CORAL will learn rank-consistent decision boundaries that explicitly preserve severity progression, allowing the model to better understand the ordered relationship between diabetic retinopathy grades. By integrating Focal weighting into the CORAL framework, the proposed loss function will simultaneously address both imbalance and ordinality, enabling the

model to prioritize difficult minority samples while preserving clinically meaningful grading structure. The following is the loss function we formulated:

$$\mathcal{L}_{\text{Focal-CORAL}} = - \sum_{i=1}^N \sum_{k=1}^{K-1} \alpha_k (1 - p_{t,i}^{(k)})^\gamma \log(p_{t,i}^{(k)})$$

$$p_{t,i}^{(k)} = \begin{cases} p_i^{(k)}, & \text{if } y_i^{(k)} = 1 \\ 1 - p_i^{(k)}, & \text{if } y_i^{(k)} = 0 \end{cases}$$

$$p_i^{(k)} = \sigma(z_i^{(k)})$$

$$y_i^{(k)} = \mathbb{I}(y_i > k), \quad k = 1, 2, \dots, K - 1$$

In addition to loss function improvements, extensive preprocessing will be applied to enhance image quality and reduce irrelevant variability in retinal fundus images. Medical retinal images often suffer from inconsistent illumination, poor contrast, noise, and imaging artifacts due to differences in acquisition devices and conditions. To standardize the dataset, Ben Graham preprocessing will be employed to normalize illumination, suppress background variation, and enhance retinal structures by subtracting local average intensity. This will improve visibility of important pathological features such as blood vessels, microaneurysms, exudates, and hemorrhages. Furthermore, Contrast Limited Adaptive Histogram Equalization (CLAHE) will be applied to improve local contrast, particularly in darker or low-quality regions, making subtle lesions more distinguishable while preventing excessive noise amplification. These preprocessing techniques will collectively improve feature clarity, reduce inter-image variability, and help the model focus on clinically relevant patterns rather than acquisition-related inconsistencies.

To further address class imbalance, data augmentation strategies will also be applied more aggressively to minority classes. Transformations such as rotation, flipping, zooming, brightness adjustments, and cropping will increase sample diversity while preserving pathological relevance. This will not only improve representation of under-sampled severity grades but also enhance model generalization by exposing it to a broader range of retinal variations. Together, the combination of hybrid ordinal-imbalance-aware loss functions and robust preprocessing will create a more clinically informed and technically optimized framework for diabetic retinopathy classification, with the goal of improving predictive accuracy across all severity levels, especially in minority and early-detection classes where accurate diagnosis is most critical.